Hugo Marquerie 10/12/2025

Derivada logarítmica de un producto finito de Blaschke

Lema 1. *Sea B el producto finito de Blaschke asociado a $\{z_k\}_{k=1}^n \subset \mathbb{D}$ y $\gamma \in \mathbb{T}$, entonces su derivada logarítmica viene dada por*

$$\frac{B'(z)}{B(z)} = \sum_{k=1}^{n} \frac{|z_k|^2 - 1}{(1 - \overline{z_k} z)(z_k - z)}.$$

***Demostración*:** Por definición tenemos que $B(z) = \gamma \prod_{k=1}^{n} \tau_{z_k}(z)$, luego

$$\frac{B'(z)}{B(z)} = \sum_{k=1}^{n} \frac{\tau'_{z_k}(z)}{\tau_{z_k}(z)} = \sum_{k=1}^{n} \frac{|z_k|^2 - 1}{(1 - \overline{z_k} z)^2} \cdot \frac{1 - \overline{z_k} z}{z_k - z} = \sum_{k=1}^{n} \frac{|z_k|^2 - 1}{(1 - \overline{z_k} z)(z_k - z)}.$$ ■

Referenciado en

- `lem-derivada-prod-finito-blaschke-no-nula-toro`

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